

TASK1

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jhansi_priya@Jhansi-Priya: ~/  × + ▾
jhansi_priya@Jhansi-Priya:~$ mkdir practical4
jhansi_priya@Jhansi-Priya:~$ cd practical4
jhansi_priya@Jhansi-Priya:~/practical4$ nano task1.c
jhansi_priya@Jhansi-Priya:~/practical4$ gcc task1.c -o task1
jhansi_priya@Jhansi-Priya:~/practical4$ ./task1
Parent Process Started
Parent PID: 509
Child 1 started. PID = 510 Parent PID = 509
Child 2 started. PID = 511 Parent PID = 509
All 3 child processes have been created.

Using wait():
Child 3 started. PID = 512 Parent PID = 509
Child 1 completed. PID = 510
wait() detected child PID 510 finished with exit status 10

Using waitpid():
waitpid() detected child PID 510 finished with exit status 10
Child 2 completed. PID = 511
waitpid() detected child PID 511 finished with exit status 11
Child 3 completed. PID = 512
waitpid() detected child PID 512 finished with exit status 12
Parent process completed.
jhansi_priya@Jhansi-Priya:~/practical4$
```

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GNU nano 7.2 task1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid[3];
    int status;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    for (int i = 0; i < 3; i++)
    {
        pid[i] = fork();
        if (pid[i] == 0)
        {
            printf("Child %d started. PID = %d Parent PID = %d\n",
                i + 1, getpid(), getppid());

            sleep(i + 1);

            printf("Child %d completed. PID = %d\n",
                i + 1, getpid());

            exit(10 + i);
        }
    }

    printf("All 3 child processes have been created.\n");
    printf("\nUsing wait():\n");

    pid_t finished = wait(&status);

    ^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
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GNU nano 7.2

task1.c

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        printf("Child %d started. PID = %d Parent PID = %d\n",
               i + 1, getpid(), getppid());

        sleep(i + 1);

        printf("Child %d completed. PID = %d\n",
               i + 1, getpid());

        exit(10 + i);
    }
}

printf("All 3 child processes have been created.\n");

printf("\nUsing wait():\n");

pid_t finished = wait(&status);

printf("wait() detected child PID %d finished with exit status %d\n",
       finished, WEXITSTATUS(status));

printf("\nUsing waitpid():\n");

for (int i = 0; i < 3; i++)
{
    waitpid(pid[i], &status, 0);

    printf("waitpid() detected child PID %d finished with exit status %d\n",
           pid[i], WEXITSTATUS(status));
}

printf("Parent process completed.\n");

return 0;
}
```

TASK2

```
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid;
    pid = fork();
    if (pid == 0)
    {
        printf("Child process: PID = %d\n", getpid());
        printf("Child process terminating...\n");
        exit(0);
    }
    else
    {
        printf("Parent process: PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
        printf("Parent sleeping...\n");
        sleep(30);
        printf("Parent terminating...\n");
    }
    return 0;
}
```

```
waitpid() detected child PID 512 finished with exit status 12
Parent process completed.
jhansi_priya@Jhansi-Priya:~/practical4$ nano task1.c
jhansi_priya@Jhansi-Priya:~/practical4$ nano task2.c
jhansi_priya@Jhansi-Priya:~/practical4$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/practical4$ ./task2
Parent process: PID = 544
Child PID = 545
Parent sleeping...
Child process: PID = 545
Child process terminating...
Parent terminating...
jhansi_priya@Jhansi-Priya:~/practical4$ nano task2.c
jhansi_priya@Jhansi-Priya:~/practical4$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/practical4$ ./task2.c
-bash: ./task2.c: Permission denied
jhansi_priya@Jhansi-Priya:~/practical4$ ./task2
Parent process: PID = 726
Child PID = 727
Parent sleeping...
Child process: PID = 727
Child process terminating...
Parent terminating...
jhansi_priya@Jhansi-Priya:~/practical4$ nano task2.c
jhansi_priya@Jhansi-Priya:~/practical4$ gcc task2.c -o task2
jhansi_priya@Jhansi-Priya:~/practical4$ ./task2.c
-bash: ./task2.c: Permission denied
jhansi_priya@Jhansi-Priya:~/practical4$ ./task2
Child process terminating...
Zombie process removed.
jhansi_priya@Jhansi-Priya:~/practical4$ |
```

```
jhansi_priya@Jhansi-Priya: ~  
jhansi_priya@Jhansi-Priya:~$ ps -el | grep task2  
0 S 1000 726 302 0 80 0 - 672 hrttime pts/0 00:00:00 task2  
1 Z 1000 727 726 0 80 0 - 0 - pts/0 00:00:00 task2  
jhansi_priya@Jhansi-Priya:~$
```

```
GNU nano 7.2 task2.c  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    pid_t pid = fork();  
  
    if (pid == 0)  
    {  
        printf("Child process terminating...\n");  
        exit(0);  
    }  
    else  
    {  
        wait(NULL);  
  
        printf("Zombie process removed.\n");  
    }  
  
    return 0;  
}
```

